

SOP-CP-3500: Cathodic Protection System Monitoring

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Section 1. Purpose and Scope

Governs monitoring and maintenance of impressed-current and sacrificial-anode cathodic protection (CP) systems on buried pipelines and tank bottoms.

Section 2. Survey Frequency

Clause 2.1 — Pipe-to-soil potential surveys are conducted at least annually at each test station, and rectifier output is checked at least every 2 months for impressed-current systems.

Section 3. Protection Criteria

Clause 3.1 — A protected structure shall read at least -850 mV (copper-sulfate reference electrode, instant-off) at every test point; a reading less negative than this threshold indicates inadequate protection requiring corrective action.

Clause 3.3 — A test point failing the protection criterion for two consecutive survey periods triggers an engineering evaluation of that structure's corrosion risk and CP system adequacy, not just continued monitoring.

Section 4. Rectifier and Anode Maintenance

Clause 4.1 — A rectifier found with zero or abnormal output at a bi-monthly check is repaired within 15 days; the affected structure's protection status during the outage is evaluated by Corrosion Engineering.

Section 5. Records

Clause 5.1 — Survey data and rectifier logs are filed to the Corrosion Engineering File and retained for the operating life of the protected structure.